

$$Q_p = \mp 1.609 * 10^{-19} C$$

$$i = \frac{dq}{dt}$$

$$\Delta q(t) = \int_{t_0}^t i dt'$$

$$P = Vi = i^2 R = \frac{V^2}{R}$$

$$[V] = v, [Q] = C, [W] = J, [i] = A$$

$$V = iR$$

$$G = \frac{i}{V} = R^{-1}$$

$$w = vQ$$

$$\sum_I i = 0 \text{ for some node I}$$

$$\sum_P V = 0 \text{ for some path P}$$

$$\sum_E P_{\text{absorbed}} = \sum_E P_{\text{supplied}}$$

for a set of elements E

$$R_{\text{eq}} = \sum R_i \text{ (series)}$$

$$\frac{1}{R_{\text{eq}}} = \sum \frac{1}{R_i} \text{ (parallel)}$$

$$v_k = \left(\frac{R_k}{R_{\text{eq}}} \right) v$$

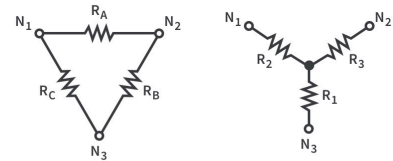
$$i_k = \left(\frac{R_{\text{eq}}}{R_k} \right) i$$

A planar circuit has no intersecting wires

$$p_L = i_L^2 R_L = \frac{v_s^2 R_L}{(R_s + R_L)^2}$$

$(R_L \text{ will absorb max power}) \rightarrow (R_L = R_s)$

$$p_{\text{max}} = \frac{v_s^2}{4R_s} = \frac{v_{\text{TH}}^2}{4R_{\text{TH}}}$$



$$R_A = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_1}$$

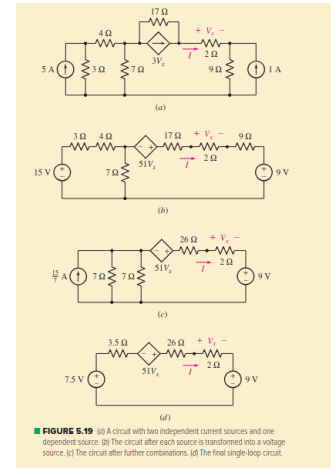
$$R_B = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_2}$$

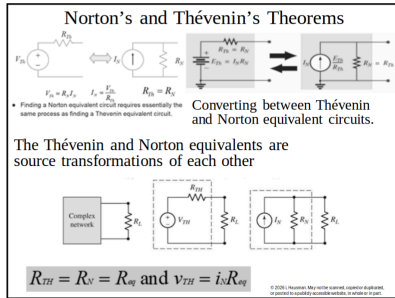
$$R_C = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_3}$$

$$R_1 = \frac{R_B R_C}{R_A + R_B + R_C}$$

$$R_2 = \frac{R_C R_A}{R_A + R_B + R_C}$$

$$R_3 = \frac{R_A R_B}{R_A + R_B + R_C}$$





No i ever flows
into an I.O.A. terminal

No volt. diff. between
I.O.A. terminals ($v_+ = v_-$)

$$i = \frac{v_{in}}{R_1}$$

$$v_{out} = \left(-\frac{R_f}{R_1} \right) v_{in}$$

TABLE 6.1 Summary of Basic Op Amp Circuits

Name	Circuit Schematic	Input-Output Relation
Inverting Amplifier		$v_{out} = -\frac{R_f}{R_1} v_{in}$
Noninverting Amplifier		$v_{out} = \left(1 + \frac{R_f}{R_1} \right) v_{in}$
Voltage Follower (also known as a Unity Gain Amplifier)		$v_{out} = v_{in}$
Summing Amplifier		$v_{out} = -\frac{R_f}{R} (v_1 + v_2 + v_3)$
Difference Amplifier		$v_{out} = v_2 + v_1$

+ is the non-inverting input
- is the inverting input